

Query-Efficient Quantum Approximate Optimization via Graph-Conditioned Trust Regions

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Abstract

Parameter transfer can provide a useful starting point for the quantum approximate optimization algorithm, but finite measurement budgets can limit subsequent refinement. We formulate a graph-conditioned trust-region policy for MaxCut that combines donor-derived geometry, sampled affine models, radius-dependent shot allocation, and statistical acceptance tests. The analysis connects local-model error to sufficient conditions for affordable acceptance and distinguishes decision validity from detection power. In ideal-circuit simulations, a 256-run experiment on 16 graphs crosses sampling allocation with stopping after an unresolved test. All four policies return identical parameters in 64 matched graph/depth/repetition cases. Relative to continuation that refits at the same radius, stopping reduces mean optimization shots by 38–42%; changing allocation produces no additional quality improvement. A separate shape-by-rule experiment accepts four of 288 proposals, with smaller mean gains for the shaped policy than for simultaneous perturbation stochastic approximation (SPSA) and constrained optimization by linear approximation (COBYLA). Conditional diagnostics on eight additional graphs show how model accuracy, proposal margin, and endpoint resolution jointly determine accepted gain and measurement cost. These results establish stopping-related savings within the tested policies and identify the conditions limiting refinement. Predictions using exact simulator inputs remain offline diagnostics; the study does not establish a general optimization or query-complexity advantage from graph-conditioned geometry.

1 Introduction

The quantum approximate optimization algorithm (QAOA) optimizes an objective over parameterized quantum states [9]. Even a shallow circuit requires repeated measurements at each candidate parameter vector. Transferring parameters from a related graph can provide a useful starting point, but further refinement must produce improvements large enough to resolve within the remaining measurement budget. We study this problem for MaxCut, separating the quality inherited from transfer from the gain and cost of subsequent optimization.

Parameter transfer is supported by interpolation and Fourier initialization [33], annealing-inspired schedules [23], concentration within specified random graph ensembles [2], local subgraph structure and graph representations [8, 10], and precomputed tree angles [27, 31]. Recent methods also generate graph-conditioned trajectories [19] or refine a selected layer after transfer [21]. These methods differ in training, preprocessing, and online measurement costs. Initialization-only controls are therefore necessary to identify the contribution of refinement.

Our proposed refinement policy combines a graph-selected initial vector, called the anchor, with a fixed donor-derived region shape, an affine interpolation model, and fresh endpoint measurements for acceptance. Radius-dependent model sampling offsets the explicit inverse-radius

factor in the gradient-noise scale; an unresolved acceptance test terminates the run. The policy, labeled “RA” (resolution-aware) in figures, is compared with its original version, which contracts and continues after an unresolved test. An isotropic control tests the chosen geometry, but separating graph-dependent orientation from anisotropy would additionally require spectrum-matched random rotations. We use circuit shots as the primary cost; an objective-estimation call and a logical measurement batch are recorded separately.

The constituent numerical ideas have substantial precedent. QR-based fitting and interpolation geometry control model sensitivity [5, 12]; local quadratic models and observation reuse already improve variational optimizers [28], and ANASTAARS uses noise-aware adaptive random subspaces [6]. Shot allocation is central to iCANS [17] and to spatial objective-and-variance modeling for QAOA [13]. Stochastic trust-region convergence theories require accuracy and sampling conditions [4, 24] that the present finite-horizon policy does not enforce. Using an empirical-Bernstein endpoint test does not reproduce those published optimizers or establish an advantage over them.

We ask which components change the returned quality or measurement cost, and when a useful model proposal can be resolved by its acceptance test. Two factorial experiments separate region shape and confidence rule from sampling allocation and unresolved stopping. Conditional diagnostics on additional graphs connect model error, proposal margin, and accepted gain. The analysis supplies explicit sufficient conditions and a policy-specific power bound; the experiments determine their practical relevance. Exact simulator inputs used for diagnosis are distinguished from information available to the implemented optimizer.

2 Graph-conditioned refinement and guarantees

For a finite simple unweighted graph $G = (V, E)$ with $|E| > 0$ at depth p , minimize the normalized objective $f_G = -F_G/|E| \in [-1, 0]$, where F_G is the expected cut and $d = 2p$. The default selector compares standardized degree and clustering descriptors to choose an anchor x_0 from an offline bank. A regularized scatter matrix of nearby, symmetry-aligned donor parameters defines a fixed symmetric positive-definite shape B , with $\det B = 1$ and $\text{cond}(BB^\top) \leq 4$; the isotropic control uses $B = I$. The graph-dependent choices precede target optimization and use no target-objective measurements. An affine model $m(u) = \hat{c}_0 + \hat{c}_{1:d}^\top u$ is fitted at $d + 1$ selected points in $x + \Delta B\{u : \|u\|_2 \leq 1\}$. The proposed direction uses the fitted gradient coefficients, whereas fresh endpoint observations determine acceptance. For a proposal u^+ , write $q = m(0) - m(u^+)$ for predicted decrease, $a = f_G(x) - f_G(x^+)$ for true improvement, and $h = a - \eta q$ for its sufficient-decrease margin. Appendix A specifies the interpolation geometry and statistical assumptions; Appendix B defines the graph selector and shape construction. Model shots are fixed before each fit by

$$S(\Delta) = \max\{2, \lceil S_0(\Delta_0/\Delta)^2 \rceil\}. \quad (1)$$

2.1 Resolution-aware refinement

The resolution-aware routine **refine** takes an anchor x_0 , a fixed shape B , a shot cap $N_{\max}$, initial radius Δ_0 , initial model count S_0 , and prescribed acceptance looks, with $d = 2p$. In the shape-by-rule experiment, $(S_0, \Delta_0) = (256, 0.1)$, $T = 16$, $\eta = 0.1$, $\alpha = 0.05$, and $s = (512, 2048, 8192, 16384)$, with radius bounds $[0.025, 0.8]$. The input shape is I or the bounded descriptor shape. Its output is the last accepted incumbent, an explicit stopping reason, and the complete measurement record. The cap covers optimization measurements; the 8192-shot independent output evaluation is additional. Original-policy controls are specified in Appendix B.

1. Initialize $x = x_0$, $\Delta = \Delta_0$, and spent shots to zero. Record the anchor as the incumbent at zero target-optimization shots; reference-bank construction remains an offline cost. Seed separate streams for classical point selection, model sampling, and endpoint sampling.
2. Before trial t , compute $S(\Delta)$ by Eq. (1). If the remaining shots cannot cover $(d+1)S(\Delta)+2s_1$, stop with **resolution_limited** and reason **model_and_first_look_unaffordable**. Spend no model shots for that untestable proposal. Reserving the first look assures feasibility of one test; it does not assure detection at that look.
3. Select a full-rank QR design, sample its $d+1$ points at $S(\Delta)$, and fit the affine model. Record radius, design, coefficients, shot count, and condition number. If $q = \|\hat{c}_{1:d}\|_2 < 10^{-12}$, return the incumbent with a flat-model resolution reason, retaining the model shots already spent; this is not a stationarity test. Otherwise set $u^+ = -\hat{c}_{1:d}/q$ and propose $x^+ = \text{wrap}(x + \Delta B u^+)$. The wrap operation applies the exact angle periodicities specified in Appendix A.
4. At each prescribed look request fresh observations at both endpoints, adding only $s_\ell - s_{\ell-1}$ samples per endpoint, with $s_0 = 0$. If the next two-endpoint batch is unaffordable, consume none of it and mark the trial unresolved for budget reasons. For an affordable look, compute the decrease estimate $\hat{a}_\ell$ and the interval $[\hat{a}_\ell - w_\ell, \hat{a}_\ell + w_\ell]$, where $w_\ell = 2b_\ell$ for Eq. (12), or the sum of the two endpoint radii for Eq. (17). Accept if its lower endpoint is at least ηq , and reject if its upper endpoint is below ηq . Otherwise continue to the next affordable look.
5. After acceptance set $x = x^+$, record its cumulative shot count, and expand Δ by 1.5 up to 0.8. Only a certified rejection contracts the radius to $\max\{0.5\Delta, 0.025\}$. A rejection when already at the radius floor terminates with **radius_limit**. An unresolved decision stops immediately with **resolution_limited**, recording whether stopping followed the final prescribed look or an unaffordable batch; it leaves both the incumbent and radius unchanged.
6. Count each attempted trial toward the declared horizon and continue unless a stopping condition applies. Return the incumbent with status **horizon** at horizon exhaustion, and return it at every other ordinary stop. Rank failures or nonfinite inputs raise explicit errors; no substitute model is used.

This policy distinguishes insufficient measurement evidence from evidence against sufficient decrease. A resolution stop means that the declared model or decision schedule has ended without resolving a further update. Other proposals, estimators, or allocations may still succeed. On the simultaneous confidence event, a certified rejection proves $h < 0$, which still permits a positive improvement $a > 0$. The original **optimize** and **bernstein_optimize** routines contract after unresolved decisions. They are used only for their explicitly labeled controls and replay.

2.2 Finite-budget guarantees

The analysis separates initialization, model accuracy, and the decision to accept a proposal. Theorem 2 bounds objective discrepancy by total variation between rooted edge-neighborhood distributions and, given an ϵ -optimal donor on a common parameter domain, bounds transferred regret. Disjoint neighborhood-type distributions give only the objective-range bound. This structural certificate does not justify the descriptor selector or the orientation of the donor-derived shape.

Proposition 3 bounds affine coefficient error by $\mathcal{E}$, separating Taylor remainder from amplified measurement noise. Proposition 4 gives the conditional bias–variance identity for the physical-gradient fitting map. For fixed design and shape, inverse-square allocation stabilizes the leading

gradient-noise scale; it neither makes that error vanish nor guarantees a useful direction. These statements motivate the radius diagnostics, rather than certify each executed model.

Theorem 5 gives the decision guarantee: with probability at least $1 - \alpha$, every accepted step in the adaptive finite horizon has true decrease at least $\eta q > 0$. Validity alone does not assure acceptance. The following sufficient condition connects model error, endpoint resolution, and the total cost of one update. For final cumulative count s_L per endpoint and L prescribed looks, write $b_L = \sqrt{\log(4TL/\alpha)/(2s_L)}$.

Corollary 1 (Sufficient model accuracy and budget for acceptance). *Under the hypotheses of Proposition 3 and Theorem 5, consider at most T trials. For $\delta \in (0, 1)$, use δ/T in each trial’s Eq. (8), with resulting bound $\mathcal{E}$ and a valid trial-region gradient-Lipschitz constant L_f . Let N_{rem} be the budget before model measurements and N_{model} their cost, equal to $(d+1)S(\Delta)$ for the implemented square design. With probability at least $1 - \delta - \alpha$, the following implication holds simultaneously at every executed nonflat trial, with $q \geq 10^{-12}$ and $u^+ = -\hat{c}_{1:d}/q$: if*

$$(1 - \eta)q \geq \mathcal{E} + \frac{1}{2}L_f\Delta^2\|Bu^+\|_2^2 + 4b_L, \quad (2)$$

$$N_{\text{model}} + 2s_L \leq N_{\text{rem}},$$

then the Hoeffding rule accepts by the final look, provided unresolved trials proceed through the prescribed looks whenever affordable.

The fitted intercept cancels in the predicted decrease, leaving one coefficient-error allowance and one Taylor remainder. The statement is a simultaneous implication on the confidence events; it is not a coverage guarantee conditional on selecting models that satisfy the random inequality. The policy estimates neither L_f nor $\mathcal{E}$ and does not enforce this sufficient certificate. Proposition 7 separately bounds the probability that the specified Hoeffding rule accepts a small-margin proposal. Full proofs and the empirical-Bernstein extension are in Appendix A. These results connect accuracy to affordable decisions without establishing a general optimization or query-complexity advantage.

3 Results

The main comparisons use unweighted MaxCut instances with 12 or 14 vertices, depths $p = 2, 3$, and ideal computational-basis measurements simulated from the exact cut distribution. Optimizer outcomes are evaluated by exact expectations at the returned parameters. Optimization-shot counts include unsuccessful attempts; each returned optimizer output also incurs 8192 independent evaluation shots, reported separately. Optimizer intervals are descriptive pointwise 95% graph-bootstrap intervals after averaging measurement repetitions within each graph, conditional on the two fixed reference banks. Diagnostic captions distinguish graph uncertainty from conditional Monte Carlo uncertainty. Appendix B gives the sampling protocols, costs, and uncertainty calculations.

3.1 Quality at a measured shot budget

We measure refinement by the gain over each policy’s own anchor, $I_G = 100[f_G(\theta_0) - f_G(\theta_{\text{out}})]$, in percentage points. The complementary signed deficit $D_G = 100[f_G(\theta_{\text{out}}) - f_G^{\text{ref}}]$ uses a best-found, uncertified reference. The shape-by-rule experiment comprises 256 runs on 16 previously examined targets, two fixed banks, depths two and three, and a 65,536-shot optimization cap. It is a post hoc comparison with matched anchors and previously selected settings. The conventional comparators are simultaneous perturbation stochastic approximation (SPSA) and constrained optimization by linear approximation (COBYLA). The shaped resolution-aware Bernstein policy gains 0.065 and

0.032 percentage points at the two depths, compared with 0.196 and 0.359 for SPSA and 0.083 and 0.197 for COBYLA. COBYLA also uses fewer mean shots. The paired gain differences from SPSA are -0.13 (95% interval -0.21 to -0.05) and -0.33 (-0.47 to -0.20) percentage points. Figure 1 retains initialization at zero shots and follows each available output through completed measurement batches. These trajectories quantify the observed cost–quality tradeoff. The paired analysis and measurement accounting are specified in Appendix B.

3.2 Stopping accounts for the savings

The separate sampling-by-stopping factorial crosses fixed or radius-dependent model shots with stopping or refitting after an unresolved test. It contains 256 shaped Bernstein runs on the same 16 targets. All four policies return identical vectors in the 64 matched graph/depth/repetition cases. With fixed model shots, stopping reduces mean expenditure by 39.76% (95% graph interval 34.65–44.02%) at depth two and 41.74% (36.17–46.43%) at depth three; the corresponding radius-dependent savings are 38.32% and 40.40%. Figure 2 attributes these savings to termination, with no returned-quality gain from the allocation intervention. The continuation control retains its radius, whereas the original policy contracts it. The shared cap also restricts continuation: two complete endpoint tests alone cost 65,536 shots, leaving nothing for their models. These comparisons therefore do not identify the best use of residual budget. The figure’s calibration panels concern a separate fresh-graph diagnostic, described below and in Appendix B.4.

3.3 Radius trades bias against noise

Figure 3 evaluates the model on the original eight-graph diagnostic panel. At fixed shot count, contraction amplifies physical-gradient noise through the radius-dependent fitting map. Increasing the shot count reduces this sampling term but leaves finite-radius affine bias. The implemented inverse-square schedule approximately stabilizes the explicit variance prefactor, while smaller radii increase model cost. Equation (11) separates these effects, using exact simulator inputs for prediction. Its numerical calibration tests the fixed linear fitting map under the ideal observation model; an online total-error prediction would additionally require measured variance information and estimates or valid bounds for model bias. Only 31 of the 256 shape-by-rule runs fit a model at a changed radius, which limits what that optimizer experiment alone can establish about the allocation schedule.

3.4 Valid tests can have little power

The 288 shape-by-rule proposals contain 178 nonimproving points, 30 improving points without positive sufficient-decrease margin, 76 positive-margin points left unresolved, and four accepted improvements. These are dependent proposals, not independent graphs. Separately, Fig. 4 fixes 160 diagnostic proposals and repeats their endpoint measurements. On its positive-margin cohort, the online four-look Bernstein rule accepts with mean frequency 0.0233 (pointwise replay interval 0.0075–0.0391) at the 32,768-shot pair cap; the mean unresolved fraction is 0.9767. The five-look extension is a different confidence schedule. Zero acceptances in 128 replays of one proposal imply a one-sided 95% upper limit of about 2.31%, so this experiment cannot validate the much smaller analytical bound in Proposition 7. The replay measures conditional power, not the success probability of a complete optimizer.

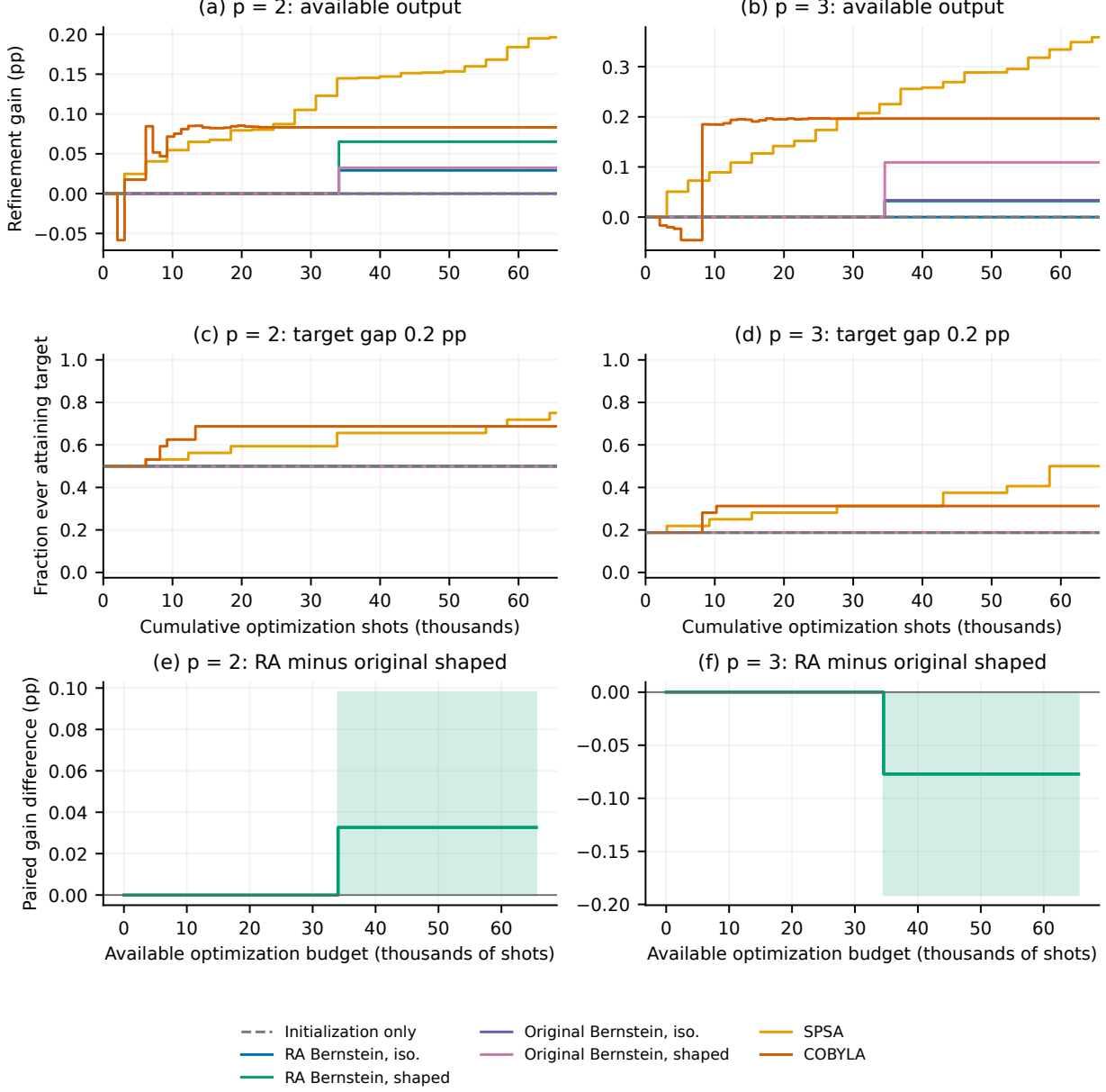


Figure 1: Solution quality versus measurement expenditure. RA denotes resolution-aware refinement. Step lines use the last recorded incumbent available at each cumulative optimization-shot budget. Until a sampling batch is complete, the preceding incumbent remains available. The anchor appears at zero shots. Target-attainment curves retain every run and count initial successes at zero shots. The target, $f_G^{\text{ref}} + 0.002$, was fixed before the resolution-aware experiment. Exact expectations evaluate recorded outputs retrospectively; they do not select them. Each trajectory comes from a run at one cap, rather than interpolation between separate runs with different caps. Pointwise 95% bands resample whole graph trajectories within the two fixed panels, each containing eight graphs with two measurement repetitions per graph and depth. Gains are in percentage points. The third row shows the paired gain difference between resolution-aware and original shaped Bernstein. Flat tails indicate that the returned output remains available after termination without further expenditure. First attainment does not imply continued satisfaction of the target for a nonmonotone baseline.

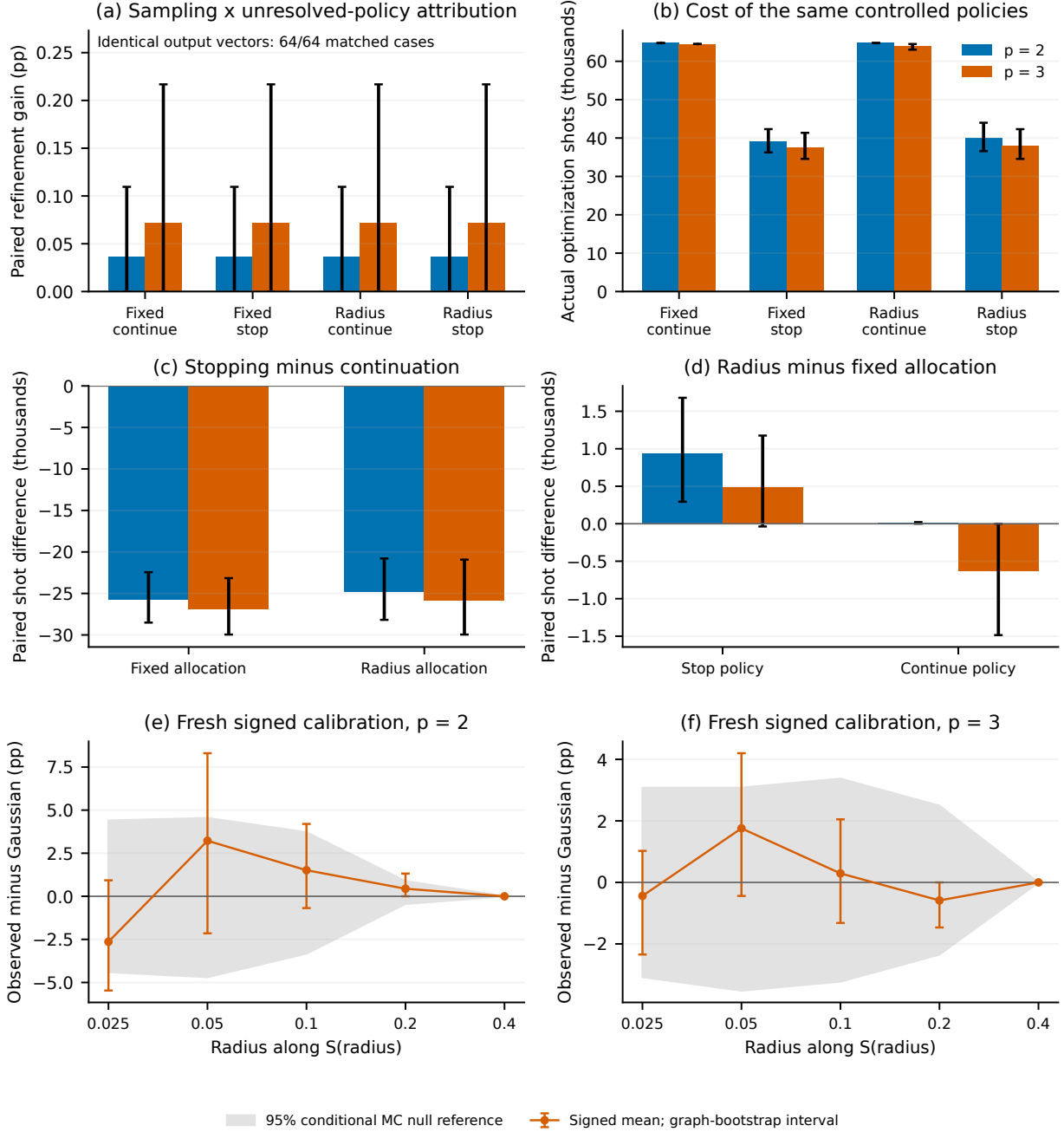


Figure 2: Attribution of expenditure and calibration residuals. Grouped bars report own-anchor gain and optimization shots for the four sampling-by-stopping policies; all four return identical parameter vectors in the 64 matched cases. Paired contrasts show stop minus continue and radius-dependent minus fixed allocation, with negative cost differences indicating savings. Continuation refits at the same radius after an unresolved test. Bottom panels show fresh-graph observed minus Gaussian-predicted positive-margin frequency. Pointwise 95% graph-bootstrap intervals describe variation across the eight targets conditional on their fixed reference bank. A separate conditional Monte Carlo reference quantifies frequency-estimation noise in 32 sampled-model and 256 Gaussian proposals per condition, assuming equal underlying probabilities within each condition. The reference is not a bound on Gaussian approximation error. All intervals are pointwise.

Shaped models: eight diagnostic graphs, depths 2 and 3 equally averaged

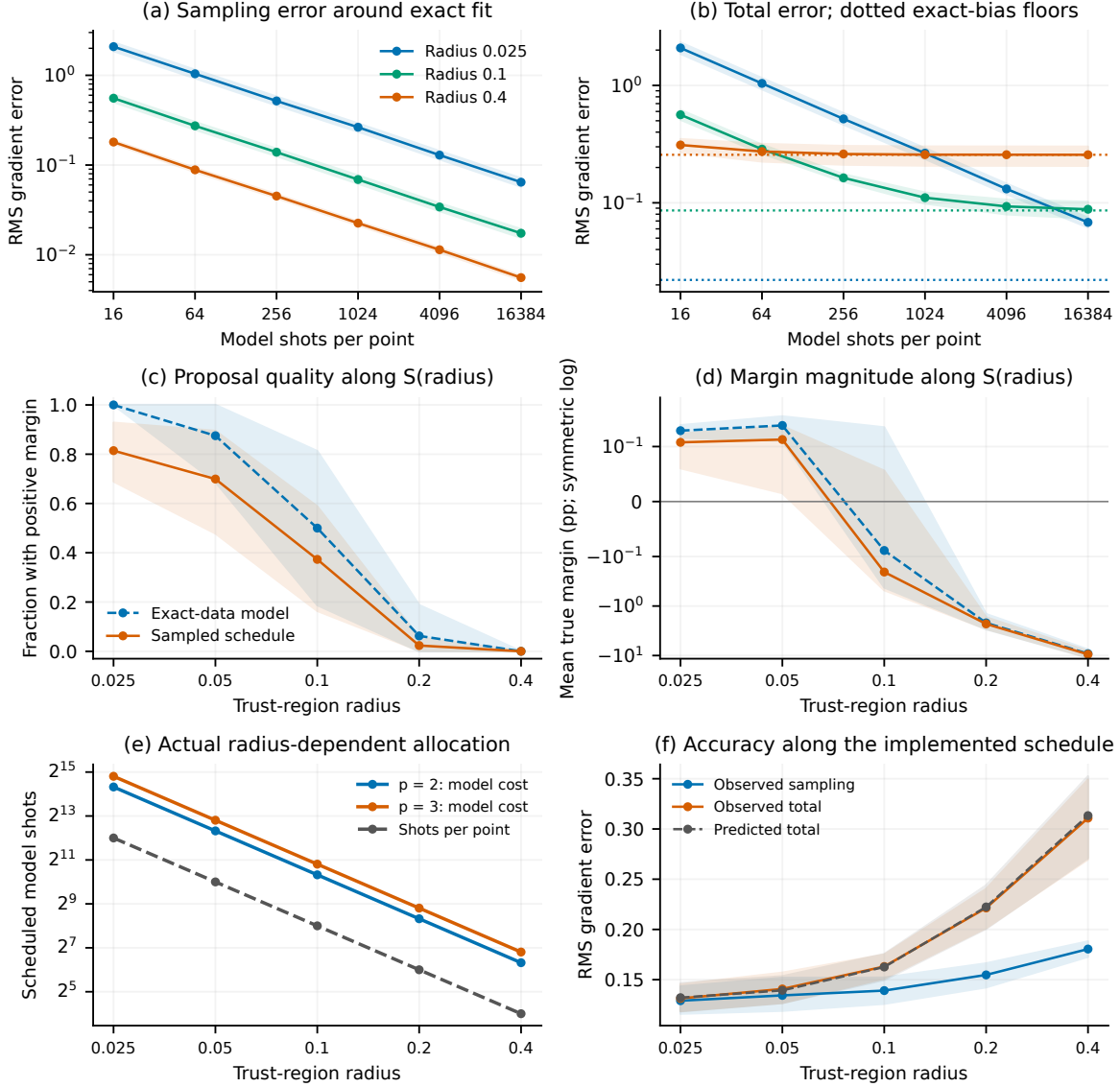


Figure 3: Radius, model bias, and propagated shot noise. Lines report root-mean-square (RMS) sampling error about an exact-data affine gradient, RMS total error against a numerically checked exact-objective gradient, and the exact-data bias floor. The plotted models use the donor-derived shape. Gradient errors are in inverse radians, and radii are in radians. Radius panels show both positive-margin frequency and true margin magnitude; the margin axis is symmetric logarithmic with a linear region around zero. The allocation panels evaluate $S(\Delta)$, including 64 and 16 shots at radii 0.2 and 0.4, and distinguish model expenditure from error. The diagnostic population comprises the eight graphs in the first follow-up panel. Depths two and three are aggregated with equal graph weights as stated in the panels. RMS panels use 128 sampled fits per condition; margin-frequency panels use 32 sampled proposals per condition. Pointwise 95% bands resample graphs rather than treating repeated fits as new graphs. Predicted errors use the conditional variance-propagation identity with exact simulator inputs.

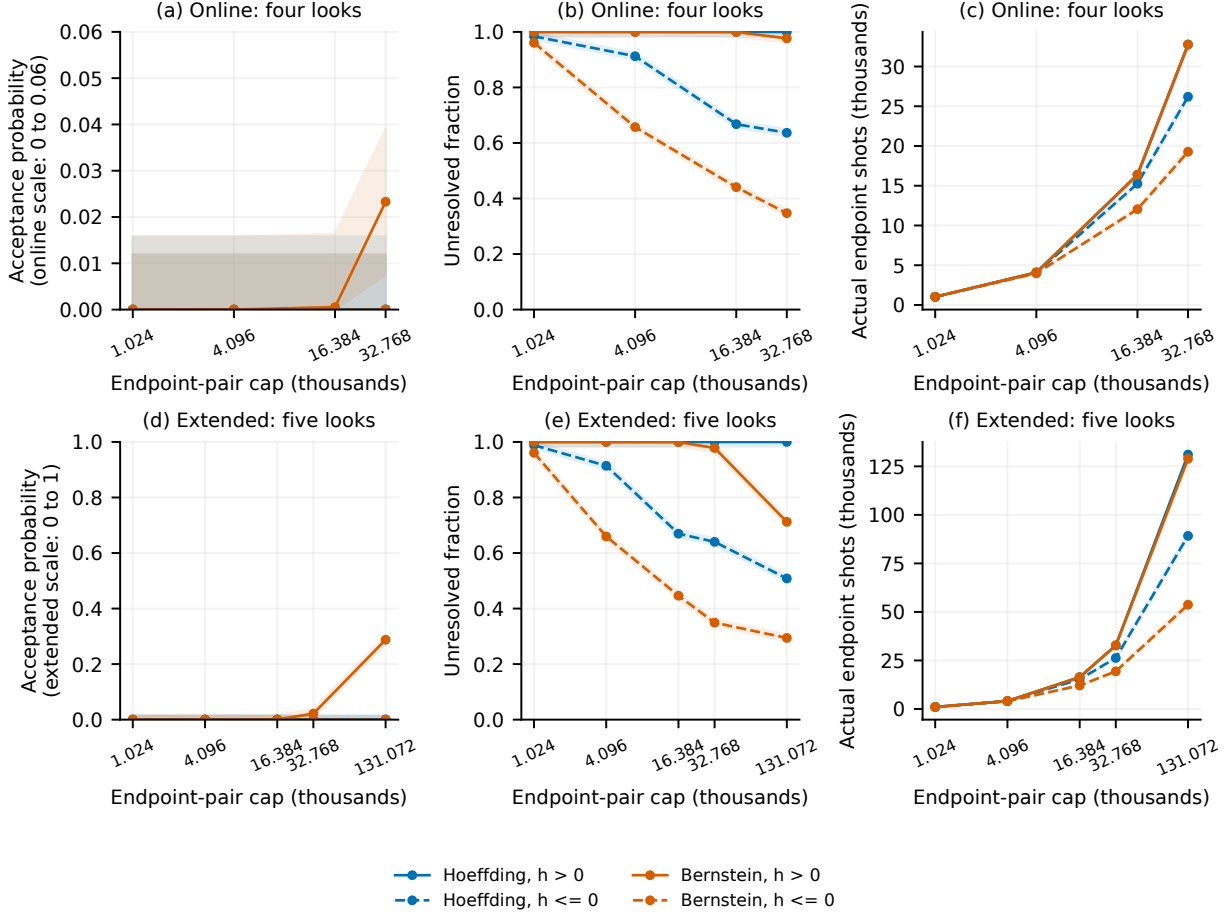


Figure 4: Acceptance power versus endpoint-pair budget. The 160 fixed diagnostic proposals use 1024 model shots per point and have equal weights within each cohort at every budget: 58 have positive margin and 102 have nonpositive margin. The curves estimate conditional replay power, rather than acceptance frequencies along optimizer trajectories. The first row uses the online four-look schedule and its $L = 4$ confidence penalty; the second uses the separately labeled five-look extension. Panels report acceptance probability, unresolved fraction, and endpoint expenditure, retaining unresolved cases at their last affordable look. Lines distinguish Hoeffding and Bernstein within fixed positive-margin and nonpositive-margin cohorts. Bands are conservative pointwise 95% Hoeffding bounds for the fixed-cohort replay means, conditional on these proposals. There are 128 replays per proposal; zero acceptances for one proposal give a one-sided 95% binomial upper limit of $1 - 0.05^{1/128} \simeq 2.31\%$. This resolution cannot establish zero power or validate a probability near 10^{-6} . Budgets are plotted on a base-two logarithmic axis. The online acceptance panel uses a 0–0.06 vertical range to resolve low power; its extension has a separate scale. This fixed- $S = 1024$ population does not implement the radius-dependent model allocation tested in Fig. 5.

3.5 From proposal margin to accepted gain

Figure 5 follows proposals on eight additional graphs through the implemented allocation and four-look decision schedule. All failed attempts remain in the gain and cost averages. At depth three, radius 0.1 gives a sampled Bernstein mean accepted gain of 0.180 percentage points (95% graph interval 0.000–0.471), concentrated on two of eight graphs, at a mean model-plus-endpoint cost of 33,336 shots. Radius 0.025 gives no accepted gain and costs 61,440 shots per attempt. Thus a high positive-margin frequency need not yield the greatest accepted gain. This diagnostic evaluates one attempted step from each fixed anchor, with repeated model and endpoint draws; it is not an optimizer benchmark or a radius-selection procedure.

The Gaussian prediction uses exact target means, variances, and trial objectives. Across all 80 fresh shaped graph/depth/radius conditions, its mean absolute positive-margin frequency difference is 2.02 percentage points (95% graph interval 1.45–2.52). This point estimate lies within the separate conditional Monte Carlo reference range, 1.65–2.78 percentage points, at the available simulation resolution. The radius/depth lookup controls have less target information, so their larger errors do not establish an online efficiency advantage. Graph intervals and Monte Carlo references answer different uncertainty questions; neither isolates Gaussian approximation error. The full protocol preserves all outcomes and charges model and endpoint measurements separately from the offline exact-simulation cost.

4 Discussion

Stopping unresolved tests reduces measurement expenditure under the specified continuation policy. Identical outputs across the sampling-by-stopping experiment exclude improved quality from radius-dependent allocation as the source of these savings. The shape-by-rule comparison shows a quality–cost tradeoff, with larger mean gains for conventional baselines. The evidence supports this specific stopping benefit without establishing general optimizer superiority.

Contraction can reduce affine bias while amplifying gradient noise at fixed shot count and shrinking the proposed improvement. Corollary 1 connects model accuracy to affordable acceptance, but the optimizer does not enforce its condition. Exact-moment calibration checks a standard linear-estimation identity; the Gaussian forecast tests the nonlinear proposal map using additional target information. Neither supplies a controller using measured inputs. The joint diagnostic concerns one attempted step from a fixed anchor.

Isotropic and shaped comparisons leave the contribution of graph-dependent orientation unresolved. Fixing anchors and eigenvalue spectra while varying orientation would separate it from initialization and anisotropy. An operational controller requires accessible estimates or valid bounds, an allocation rule, and charged pilot measurements. Relevant comparators include spatial objective-and-variance modeling [13] and time-uniform confidence sequences [16], which change model construction and sampling or replace prescribed looks. Changing the endpoint rule alone does not reproduce either approach.

The 16 optimizer targets were previously examined; eight additional graphs test a conditional mechanism with a fixed bank. Measurement repetitions are not independent graphs, and banks are not crossed with common targets. Small-graph, ideal-circuit experiments leave device noise, drift, routing, and hardware latency untested. The finite horizon, positive radius floor, and absence of fully linear model guarantees preclude general convergence or query-complexity claims. The open question is whether a practical graph-conditioned controller can produce affordable, reproducible improvements across independent problem regimes.

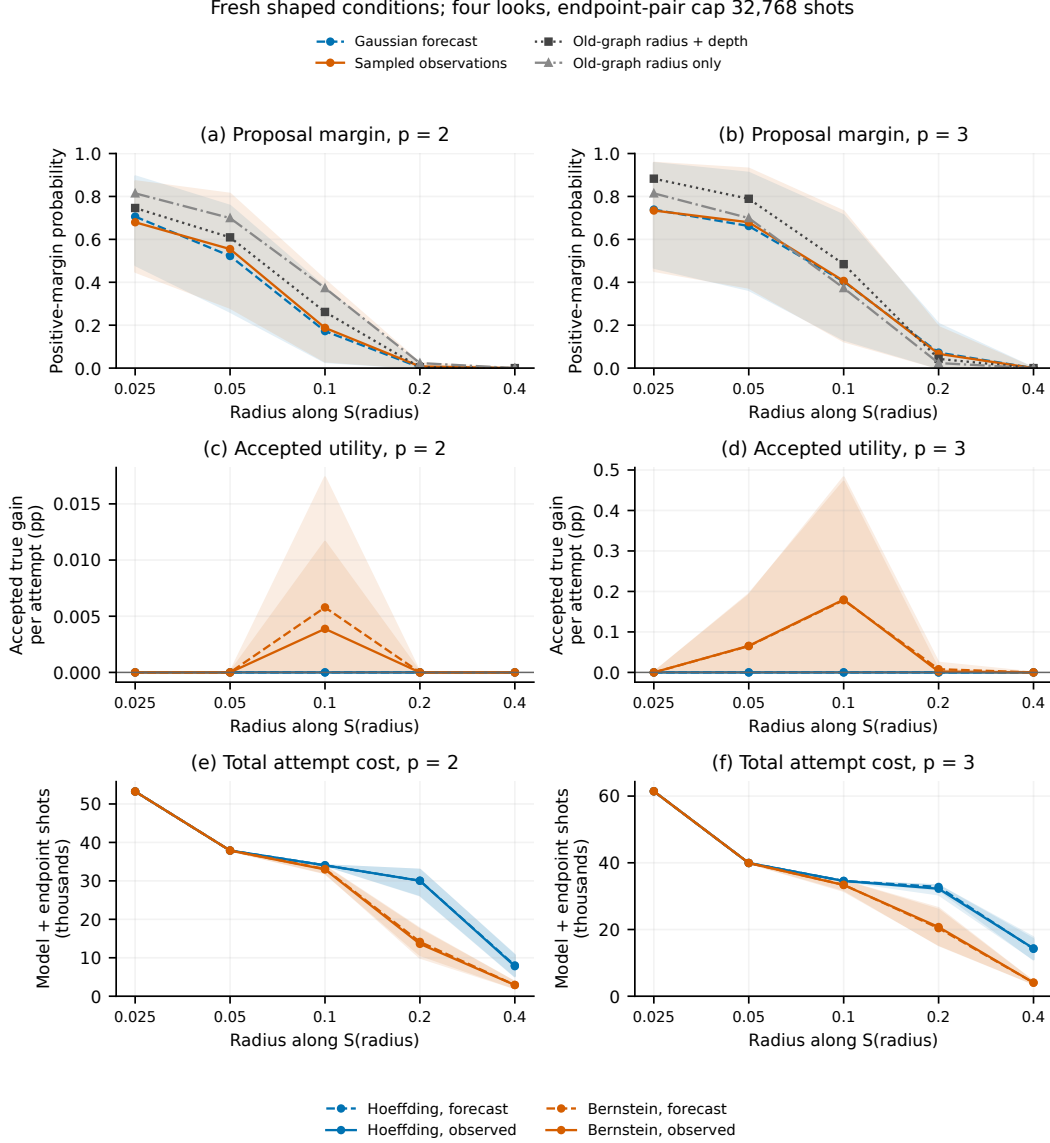


Figure 5: Proposal quality and decision cost on the same allocation schedule for eight fresh graphs with a fixed reference bank. Columns show depths two and three. Rows show positive-margin probability, accepted true gain including zero outcomes, and model-plus-endpoint shots versus radius. Scheduled per-point counts are 4096, 1024, 256, 64, and 16 at radii 0.025, 0.05, 0.1, 0.2, and 0.4. Top panels include radius-only and radius-plus-depth predictors fitted to the eight original graphs; their training set is fixed. Information access is unequal: the Gaussian forecast uses exact fresh-target moments and trial objectives, making it an offline diagnostic. Each fresh shaped condition supplies 32 sampled proposals with eight endpoint replays each, versus 256 Gaussian proposals with one replay each. Both confidence rules share endpoint streams, the online $L = 4$ penalty, and a 65,536-shot cap. Lines average graphs equally; pointwise 95% bands resample whole graphs for observed and Gaussian curves. The fixed lookup curves have no bands. The probability curves average predictions before comparison, whereas Eq. (20) averages absolute errors within individual conditions. Every proposal remains in the denominator, including nonimproving, rejected, and unresolved attempts.

A Mathematical guarantees

A.1 Objective, observations, and local coordinates

For a finite simple unweighted graph $G = (V, E)$ with $n = |V|$ and $m = |E| > 0$, define $C_G = \sum_{(i,j) \in E} (I - Z_i Z_j)/2$ and $H_X = \sum_{i \in V} X_i$. At integer depth $p \geq 1$, with $\theta = (\gamma_1, \dots, \gamma_p, \beta_1, \dots, \beta_p)$,

$$|\psi_G(\theta)\rangle = e^{-i\beta_p H_X} e^{-i\gamma_p C_G} \dots e^{-i\beta_1 H_X} e^{-i\gamma_1 C_G} |+\rangle^{\otimes n}, \quad f_G(\theta) = -\langle C_G \rangle_\theta / m. \quad (3)$$

One computational-basis observation is $Y = -C_G(z)/m \in [-1, 0]$, with mean $f_G(\theta)$. The ideal simulator draws batches from the exact cut-value distribution and charges every represented measurement. The analysis assumes fresh conditionally independent shots at each sampled point; the empirical-Bernstein extension additionally assumes identical conditional distributions within each endpoint stream.

The objective has coordinate periods 2π for γ_j and $\pi/2$ for β_j . The former follows from the integer cut spectrum; the latter from the global bit flip commuting with the circuit and preserving the initial state up to phase. Simultaneous reversal of all angles preserves the objective by complex conjugation. The operation wrap uses the coordinate periods. Local models use the unwrapped region

$$\mathcal{T}(x, \Delta, B) = \{x + \Delta B u : \|u\|_2 \leq 1\}, \quad \Delta > 0, \quad B = B^\top \succ 0, \quad \det B = 1, \quad (4)$$

with $d = 2p$. Wrapping leaves its objective values unchanged. The fixed shape preserves region volume at a given radius; this does not imply useful alignment with the objective.

A.2 Transfer through rooted neighborhoods

Let $\mu_{G,p}$ be the distribution of rooted isomorphism types obtained by sampling an edge uniformly and taking the induced subgraph on vertices within distance p of either endpoint, marked as an unordered root pair.

Theorem 2 (Transfer through rooted edge neighborhoods). *For two finite simple unweighted graphs with at least one edge at a shared depth p , let $d_p(G, H) = \text{TV}(\mu_{G,p}, \mu_{H,p})$, where $\text{TV}(\mu, \nu) = \frac{1}{2} \sum_\tau |\mu(\tau) - \nu(\tau)|$. Then*

$$\sup_{\theta \in \mathbb{R}^{2p}} |f_G(\theta) - f_H(\theta)| \leq d_p(G, H). \quad (5)$$

If $\Omega \subseteq \mathbb{R}^{2p}$ is nonempty, $\theta_H \in \Omega$, and $f_H(\theta_H) \leq \inf_{\theta \in \Omega} f_H(\theta) + \epsilon$ for $\epsilon \geq 0$, then

$$0 \leq f_G(\theta_H) - \inf_{\theta \in \Omega} f_G(\theta) \leq \min\{1, \epsilon + 2d_p(G, H)\}. \quad (6)$$

Proof. Propagate $C_e = (I - Z_u Z_v)/2$ backwards through the depth- p circuit. A mixer cannot enlarge support. Within each commuting cost layer, every gate disjoint from the pre-layer support cancels; the remaining gates add only immediate neighbors. Thus the expectation depends only on the radius- p edge neighborhood [2, 9]. Extra outer-boundary edges may be irrelevant, so this representation is sufficient rather than minimal. Root-preserving isomorphism, uniform angles, and the product initial state give a common response $h_p(\tau, \theta) \in [0, 1]$ with $-f_G(\theta) = \sum_\tau \mu_{G,p}(\tau) h_p(\tau, \theta)$. The total-variation bound for expectations of $[0, 1]$ -valued functions proves Eq. (5). Furthermore, $f_G(\theta_H) \leq f_H(\theta_H) + d_p$ and $\inf_\Omega f_H \leq \inf_\Omega f_G + d_p$. The assumed donor gap gives $\epsilon + 2d_p$; membership in Ω and the objective range give the lower bound zero and upper bound one. $\square$

Disjoint type distributions have $d_p = 1$, yielding only the objective-range bound. Multistart local minimization does not certify the donor gap ϵ . This certificate concerns rooted-neighborhood TV, not the descriptor selector, global transport score, or donor-derived shape.

A.3 Affine model accuracy and propagated variance

For local points u_i , let $A_i = (1, u_i^\top)$ and $m(u) = \hat{c}_0 + \hat{c}_{1:d}^\top u$. The sample means y_i use S_i observations, with $W = \text{diag}(S_i)$ and thin factorization $W^{1/2}A = QR$. A full-rank weighted fit is $\hat{c} = R^{-1}Q^\top W^{1/2}y$. Writing $y = Ac^* + r + e$ gives

$$\hat{c} - c^* = R^{-1}Q^\top W^{1/2}(r + e). \quad (7)$$

Here c^* contains Taylor coefficients, r is the Taylor remainder at the design points, and e is sample-mean error. The implemented square $N = d + 1$ design interpolates y : its coefficient map is A^{-1} , regardless of positive weights. QR solves evaluate the fit without forming normal equations.

The implemented design draws $d + 1$ independent standard Gaussian vectors and normalizes them to the unit sphere, then adds 0 and the signed coordinate vectors $\pm e_j$. Column-pivoted QR [3] of the transposed candidate design selects $d + 1$ rows. Only selected points incur measurements; no objective observation enters selection. A full-rank check precedes fitting. We do not establish a uniform quantitative lower bound on $\sigma_{\min}(R)$ or optimality of the selected set.

Proposition 3 (Local model error with bounded observations). *Suppose ∇f is L_f -Lipschitz on a convex set containing $\mathcal{T}(x, \Delta, B)$. Choose $N \geq d + 1$ points with $\|u_i\|_2 \leq 1$ and integers $S_i \geq 1$ before drawing independent observations in $[-1, 0]$ with mean $f(x + \Delta Bu_i)$ at point i . Assume $W^{1/2}A = QR$ has full column rank, and set $c^* = (f(x), \Delta B^\top \nabla f(x))^\top$, $a_i = L_f \Delta^2 \|Bu_i\|_2^2/2$, and $\sigma = \sigma_{\min}(R) > 0$. For $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\|\hat{c} - c^*\|_2 \leq \frac{\sqrt{\sum_i S_i a_i^2} + \sqrt{\frac{N}{2} \log(2N/\delta)}}{\sigma} =: \mathcal{E}, \quad (8)$$

and, on the same event, uniformly for $\|u\|_2 \leq 1$,

$$|f(x + \Delta Bu) - m(u)| \leq \frac{1}{2} L_f \Delta^2 \|B\|_2^2 + \sqrt{2} \mathcal{E}. \quad (9)$$

The statements hold conditionally when the region, design, and shot counts are chosen from the preceding history and the sampling assumptions hold conditional on that history.

Proof. Taylor's theorem gives $|r_i| \leq a_i$. Hoeffding's inequality [15] and a union bound give $|e_i| \leq \sqrt{\log(2N/\delta)/(2S_i)}$ for all i with probability at least $1 - \delta$. Consequently, $\|W^{1/2}r\|_2 \leq \sqrt{\sum_i S_i a_i^2}$ and $\|W^{1/2}e\|_2 \leq \sqrt{N \log(2N/\delta)/2}$. Equation (7) and $\|R^{-1}Q^\top\|_2 = 1/\sigma$ prove Eq. (8). At an arbitrary u , the Taylor remainder is at most $L_f \Delta^2 \|B\|_2^2/2$ and $\|(1, u^\top)\|_2 \leq \sqrt{2}$, proving Eq. (9). Conditioning fixes the region and sampling choices before the observations, so the same argument applies. $\square$

Let $\mathcal{H}$ contain all choices preceding the model measurements. Conditionally, put $\bar{y}_i = f(x + \Delta Bu_i)$ and $v_i = \text{Var}(Y_i | \mathcal{H})$. Independent streams with independent shots of variance v_i within point i give $\mathbb{E}[e | \mathcal{H}] = 0$ and $\Sigma_y = \text{diag}(v_i/S_i)$. With $P = (0, I_d)$, define

$$L_A = \Delta^{-1} B^{-\top} P R^{-1} Q^\top W^{1/2}, \quad \hat{g} = L_A y, \quad \bar{g} = L_A \bar{y}, \quad g = \nabla f(x). \quad (10)$$

Proposition 4 (Conditional gradient mean-square error). *Under the stated conditional unbiasedness and finite-variance assumptions,*

$$\mathbb{E}[\|\hat{g} - g\|_2^2 | \mathcal{H}] = \|\bar{g} - g\|_2^2 + \text{tr}(L_A \Sigma_y L_A^\top). \quad (11)$$

Replacing g with $\bar{g}$ leaves only the trace term.

Proof. Expand $\hat{g} - g = (\bar{g} - g) + L_A e$. Conditional unbiasedness eliminates the cross term, and $\mathbb{E}[e^\top L_A^\top L_A e \mid \mathcal{H}] = \text{tr}(L_A \Sigma_y L_A^\top)$. $\square$

This is the standard identity for a fixed linear estimator. Exact simulator moments supply its offline diagnostic inputs. For equal counts and fixed design and shape, gradient sampling error scales as $\Delta^{-1} S^{-1/2}$: Eq. (1) stabilizes this leading scale, whereas $S \propto \Delta^{-4}$ is sufficient for an $O(\Delta)$ noise bound, apart from confidence factors. Neither statement removes Taylor bias or guarantees a useful direction. Full rank alone supplies no uniform design bound, and the implementation does not estimate L_f or enforce a fully linear model condition [4, 5, 24].

A.4 Independent acceptance and finite-budget detection

For a nonflat model, $q = \|\hat{c}_{1:d}\|_2 \geq 10^{-12}$, $u^+ = -\hat{c}_{1:d}/q$, and $x^+ = \text{wrap}(x + \Delta B u^+)$. Thus $q = m(0) - m(u^+)$, $a = f_G(x) - f_G(x^+)$, and $h = a - \eta q$. Fix integers $T, L \geq 1$, $0 < \alpha, \eta < 1$, and cumulative integer looks $1 \leq s_1 < \dots < s_L$ before the run. Fresh endpoint streams give

$$b_\ell = \sqrt{\frac{\log(4TL/\alpha)}{2s_\ell}}, \quad \hat{a}_\ell = \bar{Y}_{x,\ell} - \bar{Y}_{+, \ell}. \quad (12)$$

The Hoeffding rule accepts if

$$\hat{a}_\ell - 2b_\ell \geq \eta q, \quad (13)$$

rejects if $\hat{a}_\ell + 2b_\ell < \eta q$, and otherwise remains unresolved. Later looks add only the difference in cumulative counts. Omitted or unaffordable paired batches consume no shots.

Theorem 5 (Simultaneous validity over prescribed looks). *Consider at most T adaptive trials. Conditional on the history before a trial's endpoint measurements, fix its center, proposal, and $q > 0$, then start fresh streams of conditionally independent observations in $[-1, 0]$ with the corresponding true objective means. At the prescribed looks, with probability at least $1 - \alpha$, every executed endpoint interval is valid and every acceptance satisfies $a \geq \eta q > 0$. Early stopping and budget-based omission of looks preserve this guarantee.*

Proof. For one endpoint and look, conditional Hoeffding failure probability is $2\exp(-2s_\ell b_\ell^2) = \alpha/(2TL)$. Union bounding two endpoints and L looks gives at most α/T per trial. Taking expectations and union bounding at most T trials gives α . Cumulative looks may share observations; their independence is unnecessary. On this event, $|\hat{a}_\ell - a| \leq 2b_\ell$, so Eq. (13) gives $a \geq \eta q$. Stopping only removes controlled events. $\square$

The guarantee is per run and requires no accurate local model. On the same event, rejection proves $h < 0$, which can still accompany $a > 0$; an unresolved outcome provides neither conclusion.

Corollary 6 (A sufficient measurement scale for detection). *Under Theorem 5, suppose unresolved trials proceed through the prescribed looks whenever affordable. A trial is accepted by the final look on the simultaneous confidence event if that look is affordable and*

$$h = a - \eta q \geq 4b_L > 0. \quad (14)$$

Equivalently, for $h > 0$, a sufficient final count per endpoint is

$$s_L \geq \frac{8 \log(4TL/\alpha)}{h^2}. \quad (15)$$

Proof. No earlier rejection is possible on the event, since $\hat{a}_\ell + 2b_\ell \geq a > \eta q$. At the last look, $\hat{a}_L - 2b_L \geq a - 4b_L \geq \eta q$. Substituting Eq. (12) gives Eq. (15). $\square$

Equation (15) is sufficient for this rule, not a necessary sample count or a statistical lower bound.

Proof of Corollary 1. Conditional model failure allowances δ/T , followed by expectations and a union bound, give simultaneous model validity with probability at least $1 - \delta$. Intersecting with the endpoint event gives $1 - \delta - \alpha$ without independence between the two events. On their intersection, Taylor expansion and $\|u^+\|_2 = 1$ give

$$a - q = (\hat{c}_{1:d} - c_{1:d}^*)^\top u^+ - r(u^+) \geq -\mathcal{E} - \frac{1}{2}L_f\Delta^2\|Bu^+\|_2^2.$$

The first premise gives $h \geq 4b_L$; the second funds the model and endpoint pair through the last look. Corollary 6 completes the proof. $\square$

Proposition 7 (Acceptance probability below the confidence allowance). *Fix the history $\mathcal{H}$ after selecting a trial and before its endpoint measurements. Suppose all observations within and across the fresh endpoint streams are conditionally independent, lie in $[-1, 0]$, and have their corresponding true objective means. For $h = a - \eta q$,*

$$\mathbb{P}(\text{accept this trial} \mid \mathcal{H}) \leq \min \left\{ 1, \sum_{\ell=1}^L \exp[-s_\ell(2b_\ell - h)_+]^2 \right\}, \quad (t)_+ = \max\{t, 0\}. \quad (16)$$

The bound permits early rejection and budget-based omission of looks.

Proof. At look ℓ , $\hat{a}_\ell - a$ is a sum of $2s_\ell$ centered independent terms with range length $1/s_\ell$. Their squared ranges sum to $2/s_\ell$, so one-sided Hoeffding gives $\mathbb{P}(\hat{a}_\ell - a \geq t \mid \mathcal{H}) \leq e^{-s_\ell t^2}$ for $t \geq 0$. Acceptance requires $\hat{a}_\ell - a \geq 2b_\ell - h$. Use this tail bound when the threshold is positive and the bound one otherwise, then union bound the looks. Extending fresh streams to unexecuted looks only adds possible acceptance events; look independence is unnecessary. $\square$

This is an upper bound for the specified Hoeffding rule, not for other estimators or acceptance policies.

Empirical-Bernstein extension. For a fixed endpoint with $s \geq 2$ conditionally i.i.d. observations, let $\hat{v}_s$ be the unbiased individual-outcome sample variance. The one-sided bound of Maurer and Pontil [18, Theorem 4], applied to $Y + 1$ and its reflection, gives

$$r_s = \sqrt{\frac{2\hat{v}_s \log(8TL/\alpha)}{s}} + \frac{7 \log(8TL/\alpha)}{3(s-1)}. \quad (17)$$

Assigning failure probability $\alpha/(4TL)$ to each endpoint and tail, then conditioning and union bounding as above, gives simultaneous validity with probability at least $1 - \alpha$. Replace $2b_\ell$ in the decrease interval by $r_{x,s_\ell} + r_{+,s_\ell}$; the same accept/reject inequalities retain their meaning. Cumulative variances include within-batch sums of squares and the between-batch correction. On this interval event, the sufficient condition in Eq. (2) replaces $4b_L$ by $2(r_{x,s} + r_{+,s})$ at an affordable look. These random widths alone do not give a deterministic detection budget.

B Experimental protocols

B.1 Graphs, anchors, and region shape

The main optimizer comparisons use the two-panel experiment specified in `code/configs/followup.json`, with seeds 2026091901 and 2026091902. Each panel contains eight bank graphs (four families at $n = 8, 10$), four validation graphs (one per family at $n = 12$), and eight targets (four families at $n = 12, 14$). The families are $G(n, 0.4)$, random 3-regular, Barabási–Albert attachment with two edges per new vertex, and Watts–Strogatz graphs with initial degree four and rewiring probability 0.35 [1, 11, 14, 26, 30]. Disconnected draws are rejected; the 40 accepted graphs are mutually nonisomorphic. These filters modify the base sampling distributions. One target and one bank graph are isomorphic to graphs in the earlier canonical study; all other graphs are nonisomorphic to its instances. No bank or validation graph matches an earlier test graph. The overlapping target is retained, not excluded after observing its results. Each target uses only its panel’s bank; panel differences cannot isolate a bank effect.

At each depth $p = 2, 3$, bank parameters and diagnostic references use six exact-expectation BFGS starts: one deterministic schedule and five uniform starts in the symmetry cell. SciPy uses two-point finite-difference gradients, tolerance 10^{-7} , and at most 120 iterations per start [20, 29]. The best recorded value is an uncertified reference; target references never guide optimization or donor selection.

The descriptor comprises degree mean, standard deviation, and quartiles at 0.25, 0.5, and 0.75, plus mean clustering. Euclidean differences are scaled by bank standard deviations floored at 0.1. From distances d_j , the resolution-aware selector takes the first bank index within 10^{-12} of the minimum. Shape weights are proportional to $e^{-d_j/\tau}$, with $\tau = \max\{\text{sd}(d_j), 10^{-3}\}$. Align each donor to the anchor using gamma periods 2π , beta periods $\pi/2$, and simultaneous angle reversal; retain the nearest representative, choosing positive sign on an exact tie. Retain donors within Euclidean distance one and renormalize their weights. For $V = 0.02I + \sum_j w_j(\theta_j - x_0)(\theta_j - x_0)^\top$, center its log eigenvalues, clip to $[-\log 2, \log 2]$, and center again. With resulting values ℓ_j and eigenvectors U , set $B = U \text{diag}(e^{\ell_j/2})U^\top$. This gives $\det B = 1$ and $\text{cond}(BB^\top) \leq 4$; it is a fixed geometric heuristic, not a probabilistic covariance. Matched optimizer comparisons reuse the saved anchors and shapes exactly, including the original selector’s floating-point tie behavior.

B.2 Matched optimization and controls

All local-model settings are those in Section 2. The preceding validation compared $S \in \{256, 1024\}$ and $\Delta \in \{0.1, 0.3\}$ at both depths and both Bernstein shapes on eight validation graphs: 128 runs. Selection used mean independent finite-shot output evaluations, not exact or test objectives. Every candidate returned the anchor; common evaluation seeds gave identical scores. Candidate order therefore selected $(S_0, \Delta_0) = (256, 0.1)$, without evidence of a quality advantage. Settings and validation records were hashed before the 448 original test runs. Neither SPSA nor COBYLA received separate tuning.

Figure 1 uses the subsequent shape-by-rule factorial: 16 targets, two depths, two shot seeds, and four resolution-aware policies (isotropic/shaped by Hoeffding/Bernstein), totaling 256 runs. Original records supply controls with matching graph, depth, anchor, and 65536-shot cap. Original local-model policies use fixed model shots and contract after unresolved evidence. Their Bernstein test is our endpoint-rule implementation, not a reproduction of a published spatial-variance optimizer such as ASTRO-DF [13]. SPSA uses Rademacher perturbations, $a_k = 0.15/(k + 10)^{0.602}$ and $c_k = 0.12/k^{0.101}$, with two perturbed estimates and one incumbent estimate per update [25].

COBYLA uses SciPy’s PRIMA implementation, initial radius 0.1 and tolerance 0.002 [22, 32]. Both use 1024 shots per estimate and return the point with the smallest observed incumbent estimate.

The separate allocation-by-stopping factorial in Fig. 2 has another 256 runs on these targets, all shaped Bernstein. It crosses fixed/radius-dependent model shots with stop/continue after an unresolved test. Continuation holds the radius and refits with fresh data; only certified rejection contracts it. Shared design, model, and endpoint seed streams match the first trial across all four cells. The cap limits later opportunities: two complete endpoint tests alone consume 65536 shots. Both factorials fix their protocols before new samples, but reuse previously examined targets and are exploratory.

B.3 Model error and fixed-proposal power

The model grid uses the first panel’s eight targets and eight fresh targets from the same families and sizes. Fresh graphs are nonisomorphic to every previous graph and one another, use the fixed first-panel bank, and receive no target-specific tuning. The grid crosses two depths, two shapes, radii (0.025, 0.05, 0.1, 0.2, 0.4) and shot counts (16, 64, 256, 1024, 4096, 16384): 1920 cells with 128 sampled fits each. Within each graph/depth/shape, the normalized design is fixed. Exact means and individual-shot variances specify Eq. (11) before independent observations. Central-difference gradients at 10^{-5} and 5×10^{-6} are checked at every anchor. RMS errors take the square root after averaging squared Euclidean errors. Figure 3 shows shaped models on the original eight graphs, averaging depths equally within graphs. This is an offline conditional-moment check, not a paid gradient-estimation procedure. At the scheduled allocation, 32 additional independent multinomial models per graph/depth/shape/radius condition provide proposal-margin diagnostics on both original and fresh targets.

Figure 4 instead fixes 160 earlier sampled proposals at 1024 model shots per point and replicate zero: eight graphs, two depths, two shapes, and five radii. Each has 128 independent endpoint replays, shared across rules and budget prefixes. The 58 positive-margin and 102 nonpositive-margin proposals retain equal weights within their cohorts at every budget. Online looks are (512, 2048, 8192, 16384) per endpoint; bounds are recomputed with $T = 16$, $L = 4$, $\alpha = 0.05$. The separate extension adds 65536 with $L = 5$, reaching a 131072-shot pair cap. Its confidence penalty differs. These curves charge endpoint shots only; an optimizer also pays for its model. Unresolved cases remain in all frequencies and costs. Conservative pointwise Hoeffding intervals quantify replay uncertainty conditional on the fixed proposals, not variation over graphs.

B.4 Scheduled proposals and joint acceptance

At the five radii above, the implemented schedule uses (4096, 1024, 256, 64, 16) shots per design point. For a square interpolation matrix A , exact design means $\bar{y}$, and individual-shot variances v_i , the offline Gaussian forecast draws

$$\tilde{c} \sim \mathcal{N}\left(A^{-1}\bar{y}, A^{-1} \text{diag}(v_i/S_i)A^{-\top}\right). \quad (18)$$

Writing $\tilde{q} = \|\tilde{c}_{1:d}\|_2$, it evaluates

$$\tilde{x}^+ = \text{wrap}(x - \Delta B\tilde{c}_{1:d}/\tilde{q}), \quad \tilde{h} = f_G(x) - f_G(\tilde{x}^+) - \eta\tilde{q}. \quad (19)$$

If $\tilde{q} < 10^{-12}$, keep $\tilde{x}^+ = x$ and $\tilde{h} = -\eta\tilde{q}$. All draws remain in the denominator. Each fresh graph/depth/shape/radius condition has 256 Gaussian draws, frozen before 32 independently sampled multinomial models. Exact target moments and trial objectives make this a simulator-informed forecast, not an implementable measurement-only controller or a fresh optimizer test.

Figure 5 uses all 80 fresh *shaped* conditions. Each of the 32 sampled proposals receives eight independent endpoint replays; each Gaussian proposal receives one. This gives 256 endpoint trials per condition and population, but only 32 independent sampled models. The rules share endpoint streams and use the online four-look schedule. Every attempt first charges $(2p + 1)S(\Delta)$ model shots, then takes only wholly affordable endpoint batches under 65536 shots. Gaussian model cost represents the corresponding sampled design; the cost of obtaining its exact forecast inputs is separate. Mean accepted gain is $100a \mathbf{1}\{\text{accepted}\}$, retaining zero for failed tests and signed gain for false acceptances. Costs include all failures. The endpoint protocol was fixed before these new observations, after inspection of the earlier studies. It reuses the fresh mechanism targets and tests single attempts from fixed anchors, not complete trajectories.

Post hoc radius-only and radius-plus-depth lookup controls average observed positive-margin frequencies over the original eight graphs, without fitting to fresh outcomes. Their training set is held fixed in uncertainty calculations; unlike the Gaussian forecast, they lack exact fresh-target information. For the 80 fresh shaped conditions, define

$$\text{MAE}_{\text{pp}} = \frac{100}{8 \cdot 2 \cdot 5} \sum_{g,p,r} |\hat{\pi}_{gpr} - \hat{\pi}_{gpr}^{\text{M}}|. \quad (20)$$

The observed frequency uses 32 models; the Gaussian frequency uses 256 draws. Both contribute simulation noise. Under equal within-condition probabilities, conditioning on their combined success count K_g gives $X_g \sim \text{Hypergeom}(288, K_g, 32)$. For each depth/radius, convolution over eight graphs gives the central 95% reference for $(9 \sum_g X_g - \sum_g K_g)/(256 \cdot 8)$ in Fig. 2. The separate overall MAE reference uses 10000 conditional draws across all 80 conditions. These references describe frequency-estimation noise under equality, not confidence bounds on Gaussian approximation error.

B.5 Accounting, statistical units, and records

The 65536-shot optimizer cap counts every model and endpoint measurement, including unsuccessful attempts. Every returned optimizer output receives 8192 independent evaluation shots outside this cap; they never select the output. Exact expectations assess quality retrospectively. At intermediate budgets, retain the last completed incumbent without interpolating through unfinished batches. First-hit curves include zero-shot anchor successes and all failures; the prespecified gaps are 0.002, 0.005, 0.01, and 0.02 above the best-found reference. First attainment need not persist for a nonmonotone baseline. Exact reference construction, structural preprocessing, and diagnostic simulations are separate costs, not free online inputs or hardware-timing estimates.

Graph-bootstrap intervals use 10000 resamples (seed 5713) and percentile endpoints 2.5 and 97.5 [7]. Optimizer summaries first average shot seeds within graphs; paired contrasts resample graph differences within each fixed bank. Percentage savings are $100(1 - \bar{N}_{\text{stop}}/\bar{N}_{\text{continue}})$, where each $\bar{N}$ is the corresponding mean cost; paired costs are resampled together before forming each bootstrap ratio. Diagnostics resample whole graphs with all their conditions and repetitions, separating original and fresh sets. Replays and repeated fits are not new graphs. These descriptive intervals are pointwise and unadjusted, conditional on the banks; they do not establish simultaneous coverage or bank-population effects.

Reproduction commands and recorded numerical dependencies accompany the code. The archives under `code/results/` include configurations, graphs, seeds, source snapshots and hashes, observations, and costs: `followup.json.gz` records tuning and original controls; `refinement.json.gz`, the shape-by-rule runs; `mechanism.json.gz`, the allocation factorial, model grid, and forecasts; and `decision.json.gz`, joint replay and calibration.

References

- [1] Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999. doi: 10.1126/science.286.5439.509.
- [2] Fernando G. S. L. Brandão, Michael Broughton, Edward Farhi, Sam Gutmann, and Hartmut Neven. For fixed control parameters the quantum approximate optimization algorithm’s objective function value concentrates for typical instances, 2018. URL <https://arxiv.org/abs/1812.04170>. arXiv:1812.04170.
- [3] Peter Businger and Gene H. Golub. Linear least squares solutions by Householder transformations. *Numerische Mathematik*, 7:269–276, 1965. doi: 10.1007/BF01436084.
- [4] Ruobing Chen, Matt Menickelly, and Katya Scheinberg. Stochastic optimization using a trust-region method and random models. *Mathematical Programming*, 169:447–487, 2018. doi: 10.1007/s10107-017-1141-8. arXiv:1504.04231.
- [5] Andrew R. Conn, Katya Scheinberg, and Luís N. Vicente. Geometry of interpolation sets in derivative free optimization. *Mathematical Programming*, 111:141–172, 2008. doi: 10.1007/s10107-006-0073-5.
- [6] Kwassi Joseph Dzahini, Jeffrey M. Larson, Matt Menickelly, and Stefan M. Wild. A noise-aware scalable subspace classical optimizer for the quantum approximate optimization algorithm. In *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 20–30. IEEE, 2025. doi: 10.1109/QCE65121.2025.00013. URL <https://doi.org/10.1109/QCE65121.2025.00013>.
- [7] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*, volume 57 of *Monographs on Statistics and Applied Probability*. Chapman & Hall, New York, 1993. ISBN 0-412-04231-2.
- [8] Jose Falla, Quinn Langfitt, Yuri Alexeev, and Ilya Safro. Graph representation learning for parameter transferability in quantum approximate optimization algorithm. *Quantum Machine Intelligence*, 6:46, 2024. doi: 10.1007/s42484-024-00178-9.
- [9] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. A quantum approximate optimization algorithm, 2014. URL <https://arxiv.org/abs/1411.4028>. arXiv:1411.4028.
- [10] Alexey Galda, Eesh Gupta, Jose Falla, Xiaoyuan Liu, Danylo Lykov, Yuri Alexeev, and Ilya Safro. Similarity-based parameter transferability in the quantum approximate optimization algorithm. *Frontiers in Quantum Science and Technology*, 2:1200975, 2023. doi: 10.3389/frqst.2023.1200975. arXiv:2307.05420.
- [11] E. N. Gilbert. Random graphs. *The Annals of Mathematical Statistics*, 30(4):1141–1144, 1959. doi: 10.1214/aoms/1177706098.
- [12] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, 4 edition, 2013. doi: 10.56021/9781421407944.
- [13] Yunsoo Ha, Sara Shashaani, and Matt Menickelly. Two-stage estimation and variance modeling for latency-constrained variational quantum algorithms. *INFORMS Journal on Computing*, 37(1):125–145, 2025. doi: 10.1287/ijoc.2024.0575. URL <https://pubsonline.informs.org/doi/10.1287/ijoc.2024.0575>.

- [14] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference*, pages 11–15, Pasadena, CA, 2008. URL <https://aric.hagberg.org/papers/hagberg-2008-exploring.pdf>.
- [15] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.
- [16] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, non-parametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991. URL <https://arxiv.org/abs/1810.08240>.
- [17] Jonas M. Kübler, Andrew Arrasmith, Lukasz Cincio, and Patrick J. Coles. An adaptive optimizer for measurement-frugal variational algorithms. *Quantum*, 4:263, 2020. doi: 10.22331/q-2020-05-11-263. URL <https://doi.org/10.22331/q-2020-05-11-263>.
- [18] Andreas Maurer and Massimiliano Pontil. Empirical Bernstein bounds and sample variance penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory*, 2009. doi: 10.48550/arXiv.0907.3740. URL <https://www.cs.mcgill.ca/~colt2009/papers/012.pdf>. arXiv:0907.3740.
- [19] Kien X. Nguyen and Ilya Safro. Graph-conditioned meta-optimizer for QAOA parameter generation on multiple problem classes, 2026. URL <https://arxiv.org/abs/2604.25275>. arXiv:2604.25275.
- [20] Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*. Springer Series in Operations Research and Financial Engineering. Springer, New York, 2 edition, 2006. doi: 10.1007/978-0-387-40065-5.
- [21] Shubham Patel and Utkarsh Mishra. Improving the efficiency of QAOA using efficient parameter transfer initialization and targeted-single-layer regularized optimization with minimal performance degradation, 2026. URL <https://arxiv.org/abs/2601.15760>. arXiv:2601.15760.
- [22] M. J. D. Powell. A direct search optimization method that models the objective and constraint functions by linear interpolation. In Susana Gomez and Jean-Pierre Hennart, editors, *Advances in Optimization and Numerical Analysis*, volume 275 of *Mathematics and Its Applications*, pages 51–67. Springer, Dordrecht, 1994. doi: 10.1007/978-94-015-8330-5_4.
- [23] Stefan H. Sack and Maksym Serbyn. Quantum annealing initialization of the quantum approximate optimization algorithm. *Quantum*, 5:491, 2021. doi: 10.22331/q-2021-07-01-491. URL <https://doi.org/10.22331/q-2021-07-01-491>.
- [24] Sara Shashaani, Fatemeh S. Hashemi, and Raghu Pasupathy. ASTRO-DF: A class of adaptive sampling trust-region algorithms for derivative-free stochastic optimization. *SIAM Journal on Optimization*, 28(4):3145–3176, 2018. doi: 10.1137/15M1042425.
- [25] James C. Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992. doi: 10.1109/9.119632.
- [26] A. Steger and N. C. Wormald. Generating random regular graphs quickly. *Combinatorics, Probability and Computing*, 8(4):377–396, 1999. doi: 10.1017/S0963548399003867.

- [27] Michael Streif and Martin Leib. Training the quantum approximate optimization algorithm without access to a quantum processing unit. *Quantum Science and Technology*, 5(3):034008, 2020. doi: 10.1088/2058-9565/ab8c2b. URL <https://doi.org/10.1088/2058-9565/ab8c2b>.
- [28] Kevin J. Sung, Jiahao Yao, Matthew P. Harrigan, Nicholas C. Rubin, Zhang Jiang, Lin Lin, Ryan Babbush, and Jarrod R. McClean. Using models to improve optimizers for variational quantum algorithms. *Quantum Science and Technology*, 5(4):044008, 2020. doi: 10.1088/2058-9565/abb6d9. URL <https://doi.org/10.1088/2058-9565/abb6d9>.
- [29] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- [30] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of ‘small-world’ networks. *Nature*, 393:440–442, 1998. doi: 10.1038/30918.
- [31] Jonathan Wurtz and Danylo Lykov. Fixed-angle conjectures for the quantum approximate optimization algorithm on regular MaxCut graphs. *Physical Review A*, 104(5):052419, 2021. doi: 10.1103/PhysRevA.104.052419. URL <https://doi.org/10.1103/PhysRevA.104.052419>.
- [32] Zaikun Zhang. PRIMA: Reference implementation for Powell’s methods with modernization and amelioration, 2023. URL <https://www.libprima.net>.
- [33] Leo Zhou, Sheng-Tao Wang, Soonwon Choi, Hannes Pichler, and Mikhail D. Lukin. Quantum approximate optimization algorithm: Performance, mechanism, and implementation on near-term devices. *Physical Review X*, 10:021067, 2020. doi: 10.1103/PhysRevX.10.021067.